Case Study 2: Golden State Warriors

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Problem Introduction

Golden State Warriors created Ticket Timing Model

Identified 3 types of customers

Objective: Create 2 ML Models

- Planners
- Last-Minute
- In-Between

Analysis of the Problem

- Problem Features
- Unique Observations
- Assumptions
- Limitations

Approach

- Regression & Classification Task
- Random Forest Regressor & Gradient Boosting Model
- Why were these models chosen?
- 2-Part Modeling Strategy

Findings: Regression Task

Table 1. GSW's Regression Results

customerID	Days Before Game	Simple	Multiple
1	28	16.3	27.0
2	1	5.0	1.9
3	3	15.1	9.0
4	7	14.0	12.7
5	21	14.0	15.5

Table 2. RFR Prediction Results

customerID	Days Before Game	RFR Results
1	28	23.4
2	1	3.0
3	3	7.2
4	7	7.0
5	21	20.6

Findings: Classification Task

Table 4. Metrics for Gradient Boosting Model classifying Customer Types

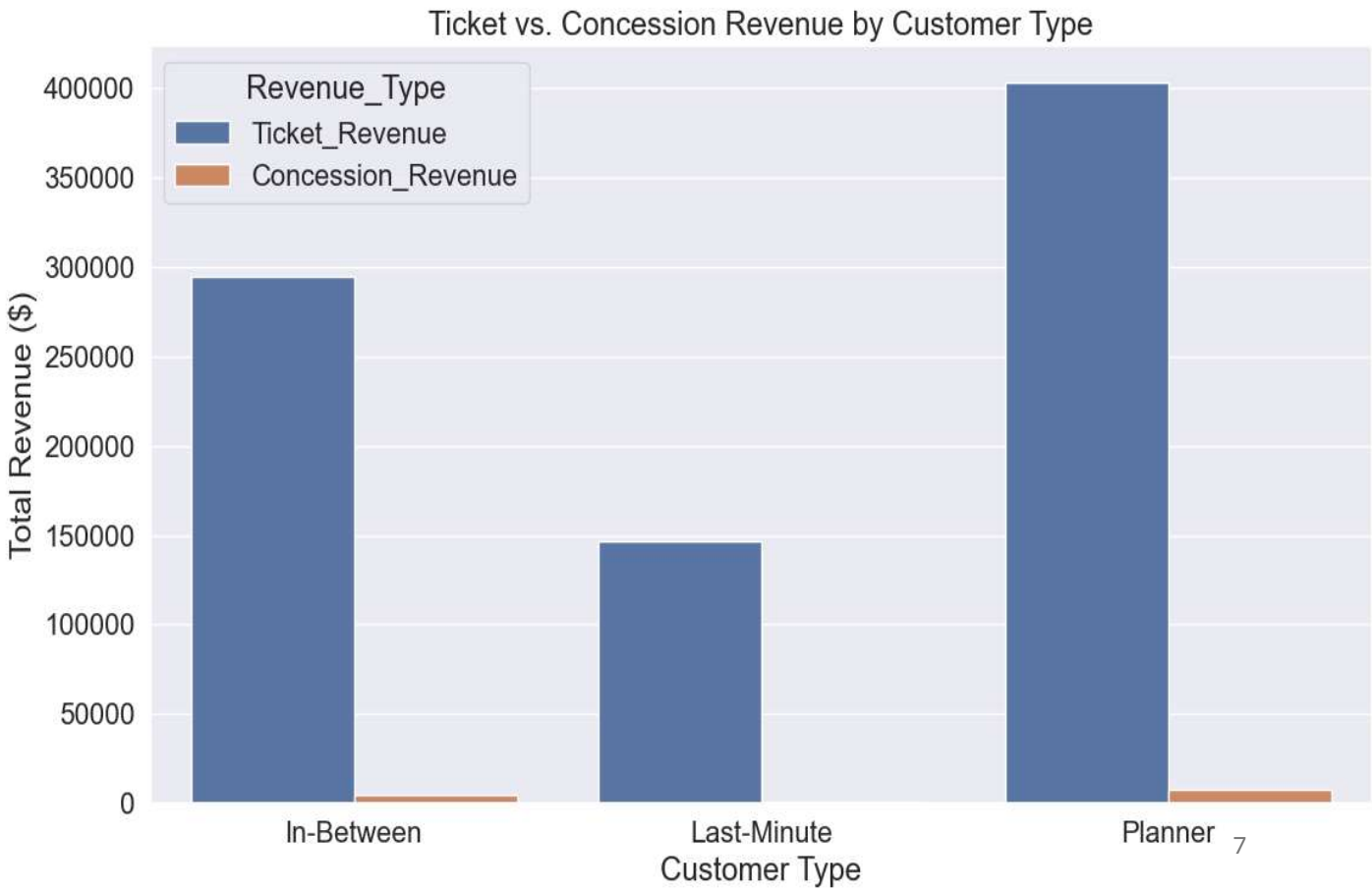
Customer type	Precision	Recall	F1-score
Last-Minute	0.93	0.88	0.90
In-Between	0.71	0.74	0.72
Planner	0.81	0.82	0.81

Table 3. Probabilities of the Decision Tree Classifying Customer Types

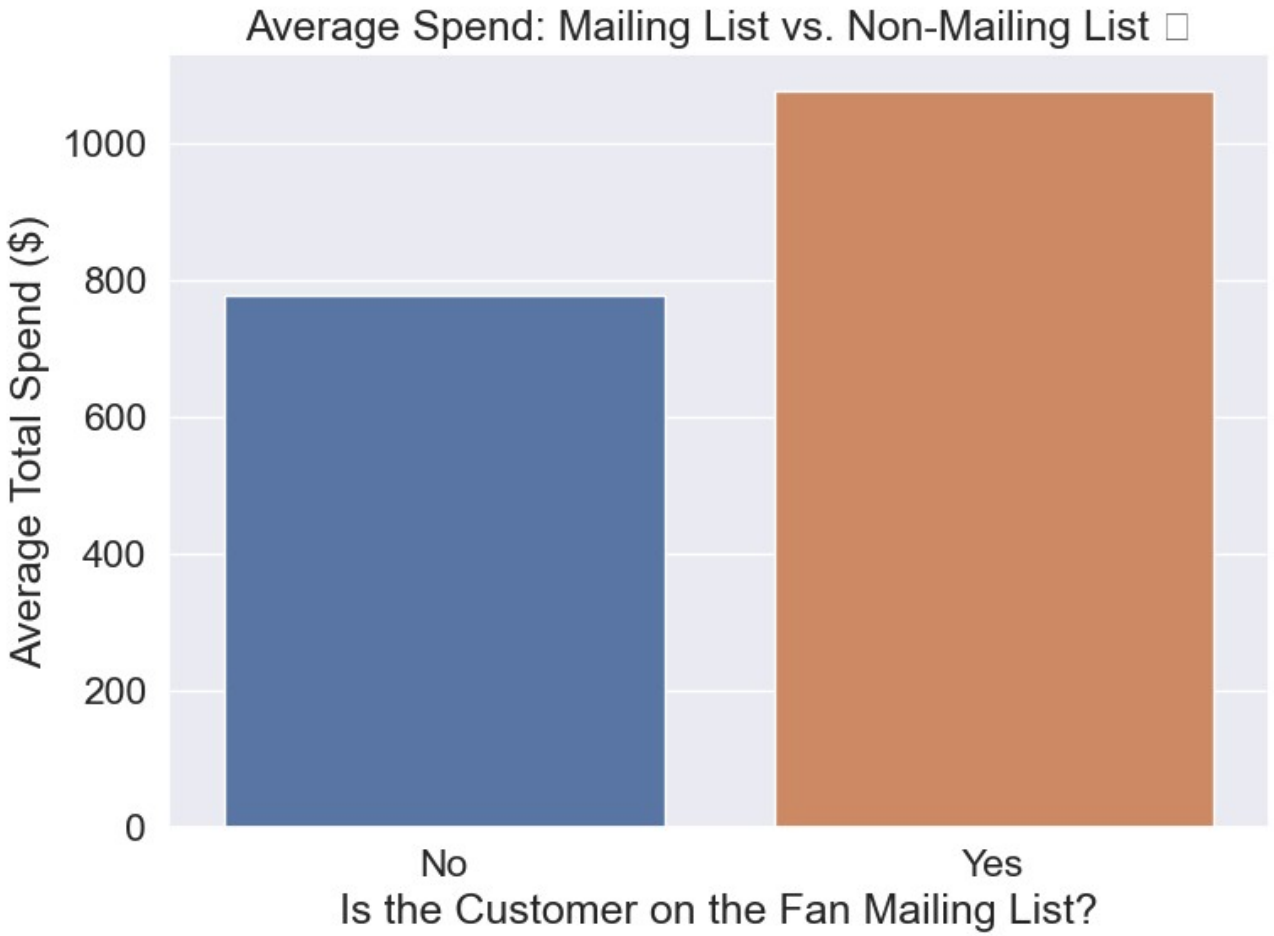
customerID	Customer Type	Last-minute probability	In-between probability	Planner probability
1	Planner	0.02	0.21	0.77
2	Last-minute	0.91	0.09	0.00
3	Last-minute	0.70	0.30	0.00
4	In-between	0.07	0.59	0.34
5	Planner	0.07	0.59	0.34

Findings: Customers and Revenue

Customer type	Ticket Revenue ($)	Concession Revenue ($)
Last-Minute	146685.06	1223.52
In-Between	294362.65	4725.53
Planner	403195.31	7570.61



Findings: Customers and Revenue



Findings: Customers and Revenue

Customer type	Lower Deck (%)	Upper Deck (%)
Last-Minute	19.57	80.43
In-Between	53.99	46.01
Planner	75.16	24.84



Verification and Validation

Verification

- Compared model performance against the GSW existing models
- No data leakage

Validation:

- Addressed business goals and provided actionable insights
- Provide clear, validated recommendations

Conclusions

- Models are an upgrade compared to the GSW's current models
- 'Planner' Customers generate the most revenue
- Addressing GSW's concerns and future data collection strategies

Recommendations



PRIORITIZE ‘PLANNER’ CUSTOMERS



SHIFT MARKETING FOCUS



ENHANCE DATA COLLECTION

Questions?

